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United States Patent	12407287
Kind Code	B2
Date of Patent	September 02, 2025
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Generator power peak limiting in virtual synchronous machine wind turbine

Abstract

Limiting generator loads in a wind turbine having a power converter with a line side inverter configured as a grid forming inverter. A maximum grid current value is determined based on a measured generator load and a limit function. A virtual impedance value and a virtual voltage drop over the virtual impedance value are determined. A virtual synchronous machine angle is determined based on a power reference and the virtual grid power. The voltage reference for controlling the line side inverter is determined based on the virtual synchronous machine angle, the virtual voltage drop and the voltage magnitude reference.

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Appl. No.:	18/723359
Filed (or PCT Filed):	December 15, 2022
PCT No.:	PCT/DK2022/050284
PCT Pub. No.:	WO2023/117004
PCT Pub. Date:	June 29, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20250062709 A1	Feb. 20, 2025

Foreign Application Priority Data

DK

PA202170656

Dec. 22, 2021

Publication Classification

Int. Cl.: H02P21/00 (20160101); F03D7/02 (20060101); H02P21/05 (20060101); H02P21/18 (20160101); H02P21/22 (20160101); H02P101/15 (20160101); H02P103/20 (20160101)

U.S. Cl.:

CPC H02P21/22 (20160201); F03D7/0284 (20130101); H02P21/05 (20130101); H02P21/18 (20160201); F05B2270/1033 (20130101); F05B2270/337 (20130101); H02P2101/15 (20150115); H02P2103/20 (20150115)

Field of Classification Search

CPC: H02P (21/22); H02P (21/05); H02P (21/18); H02P (2101/15); H02P (2103/20); F03D (7/0284); F05B (2270/1033); F05B (2270/337)

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Background/Summary

FIELD OF THE INVENTION

(1) The invention relates to control of power generation in wind turbines, and particularly control of wind turbines configured to exhibit a virtual synchronous generator response.

BACKGROUND OF THE INVENTION

(2) In order to allow a much higher penetration of renewable energy sources such as wind turbines into the electrical grid, some countries propose requirements to equip the power converters with grid-forming properties similar to conventional synchronous generators. These requirements can be address by configuring the renewable power generating units as virtual synchronous machines VSM.

(3) When the VSM configured wind turbine is operated with a high overcurrent threshold a grid disturbance such as a phase jump can lead to high peak in the generator power and/or generator torque and consequently an undesired mechanical peak load.

(4) Accordingly, it is a problem that wind turbines configured as synchronous machines may experience high mechanical loads in response to grid disturbances. Hence, an implementation of the virtual synchronous machine which addresses this and other problems is needed.

SUMMARY OF THE INVENTION

(5) It is an object of the invention to improve control wind turbines configures as virtual synchronous machines to alleviate the above mentioned problems with wind turbine loads.

(6) In a first aspect of the invention, a method for limiting generator power or generator torque levels in a wind turbine comprising a power converter with a line side inverter arranged to supply power to a power grid, the method comprises determining a maximum grid current value based on a measured generator load and a limit function which sets an upper maximum grid current value and a lower maximum grid current value so that the maximum grid current value approaches the lower maximum current value if generator load is greater than a maximum generator load and so that the maximum grid current value approaches the upper maximum current value if the generator load is smaller than the maximum generator load or is smaller than a low maximum generator load, determining a virtual impedance value based on a current difference between a current magnitude which relates to a grid current and the maximum grid current value, where the virtual impedance value comprises a virtual resistance value and optionally a virtual reactance value, determining a virtual voltage drop over the virtual impedance value based on the virtual impedance value and the grid current, determining a virtual grid power based on the virtual resistance value and the grid current, or based on the grid current and a voltage magnitude reference for a desired grid voltage or a desired reactive power to be generated by the line side inverter, determining a virtual synchronous machine angle based on a power reference for a desired power output of the power generating unit and the virtual grid power according to the virtual synchronous machine control scheme, determining a voltage reference for controlling the line side inverter to generate power based on the virtual synchronous machine angle, the virtual voltage drop and the voltage magnitude reference.

- (7) The method is particularly advantageous for wind turbines wherein the line side inverter is controlled according to a grid forming inverter scheme, i.e. wherein the virtual synchronous machine angle θ_{VSM} is determined according to the grid forming inverter scheme such as a virtual synchronous machine scheme.
- (8) By limiting the maximum grid current value to the lower maximum current value when the generator load is greater than a maximum generator load, peaks in the generator load is reduced, at least in comparison with a solution where the maximum grid current value is not limited.
- (9) By limiting the maximum grid current value to be between the lower maximum current value, which may be greater than zero, and the upper maximum current value the determination of a very small threshold value or a very large threshold value is avoided.
- (10) According to an embodiment, determining the maximum grid current value comprises determining a load difference between the measured generator load and the maximum generator load, integrating the load difference and determining the maximum grid current value based on the integrated load difference subject to the upper maximum grid current value and the lower maximum grid current value of the limit function.
- (11) Advantageously, by determining the maximum grid current value based on integrating the load difference, changes between the lower and upper maximum grid current values occurs gradually and wherein the rate of change depends on the magnitude of the load difference.
- (12) According to an embodiment, integrating the load difference comprises multiplying the load difference with a gain factor which depends on the load difference.
- (13) Advantageously, by use of an adaptive gain factor the rate of change can be adjusted. For example, by determining the gain factor dependent on the sign of the load difference, a fast change towards the lower maximum grid current can be achieved to ensure that the generator load peak will be limited, while a slower change towards the upper maximum grid current is achieved, e.g. for stability purposes.
- (14) According to an embodiment, determining the maximum grid current value comprises selecting the lower maximum grid current value if the measured generator load is greater than the maximum generator load and selecting the upper maximum grid current value if the measured generator load is smaller than the low maximum generator load, and determining the maximum grid current value by limiting a rate of change of changes between the upper maximum grid current value and the lower maximum grid current value.
- (15) According to this embodiment, the maximum grid current value is determined from a hysteresis function where the threshold values are given by the maximum generator load and the low maximum generator load. To avoid instant changes in the maximum grid current value, a rate limiter may be used to limit the rate of change.
- (16) According to an embodiment, the virtual synchronous machine angle θ_{VSM} is determined so that the double derivative of the angle θ_{VSM} is indicative of a deviation between a power reference P_{ref} and the virtual grid power P_{vsm} combined with a damping power P_d .
- (17) According to an embodiment, the virtual resistance value is determined based on a control algorithm, where the control algorithm comprises a proportional part which determines a value which is proportional with the current difference and/or an integral part which determines a value which is proportional with a time integral of the current difference.
- (18) Advantageously, by determining the virtual resistance value of the virtual impedance based on a control algorithm the virtual resistance is based on the magnitude of the current difference and possibly dependent on the integration of the error over time. This method may provide a stable determination of the virtual resistance.
- (19) According to an embodiment, the integral part comprises an integral gain which is determined dependent on the current difference, such as the sign of the current difference.
- (20) Advantageously, the integral gain could be large for positive current errors to provide fast current limitation. For negative current errors, a smaller gain can be used to provide a stable

response under unbalanced cases. In an example, the method comprises limiting the virtual resistance value according to minimum and maximum limits

(21) According to an embodiment, the virtual reactance value of the virtual impedance value is determined dependent on a predetermined factor defining a ratio between the virtual reactance value and the virtual resistance value.

(22) According to an embodiment, the current magnitude is determined as the maximum current magnitude of the current magnitudes of the current phases of the grid current, or as the magnitude of the current vector of the grid current or a transformation thereof.

(23) According to an embodiment, the virtual synchronous machine rotational speed is determined based on a combination of a feedback of a damping power, the power reference, the virtual grid power and an inertial integration model, where the synchronous machine angle is determined based on an integration of the synchronous machine rotational speed, and where the damping power is determined based on the virtual synchronous machine rotational speed.

(24) According to an embodiment, the method comprises obtaining a network voltage at a connection point of the output of the power generating unit, determining a rotational speed of the grid based on the network voltage, and determining the damping power based on the virtual synchronous machine rotational speed and the rotational speed of the grid.

(25) According to an embodiment, the method comprises determining a modified voltage reference by combining the voltage magnitude reference, or a transformation thereof, with the virtual voltage drop, and determining the voltage reference based on the modified voltage reference.

(26) The voltage magnitude reference may be defined in a DQ frame.

(27) According to an embodiment, the method comprises transforming the voltage magnitude reference to a transformed voltage magnitude reference in non-rotating target frame, wherein the modified voltage reference is determined by combining said transformed voltage magnitude reference with the virtual voltage drop.

(28) A second aspect of the invention relates to a wind turbine control system arranged for limiting generator power or generator torque levels in a wind turbine comprising a power converter with a line side inverter arranged to supply power to a power grid, wherein the wind turbine control system is arranged to perform the method of the first aspect.

(29) A third aspect of the invention relates to a wind turbine comprising a control system according to the second aspect.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Embodiments of the invention will be described, by way of example only, with reference to the drawings, in which

(2) FIG. 1 shows a wind turbine,

(3) FIG. 2A shows an example of a power system of a wind turbine or a power generating unit,

(4) FIG. 2B shows control components arranged for controlling the generation of active power and reactive power supplied to the grid at the power output of the wind turbine or power generating unit,

(5) FIG. 3A and FIG. 3B show examples of control systems for determining the synchronous machine angle of the virtual synchronous generator,

(6) FIG. 4 and FIG. 5 show examples for implementing the virtual synchronous generator with the current limiting method, and

(7) FIGS. 6A and 6B show examples of determining the maximum grid current value adaptively dependent on the generator load and a maximum generator load, and

(8) FIG. 7 illustrates simulation results showing how the adaptive determination of the maximum

grid current value limits the peak generator load.

DETAILED DESCRIPTION

(9) FIG. 1 shows a wind turbine **100** (WTG) comprising a tower **101** and a rotor **102** with at least one rotor blade **103**, such as three blades. The rotor is connected to a nacelle **104** which is mounted on top of the tower **101** and being adapted to drive a generator situated inside the nacelle via a drive train. The rotor **102** is rotatable by action of the wind. The wind induced rotational energy of the rotor blades **103** is transferred via a shaft to the generator. Thus, the wind turbine **100** is capable of converting kinetic energy of the wind into mechanical energy by means of the rotor blades and, subsequently, into electric power by means of the generator. The generator is connected with a power converter which comprises a generator side converter and a line side inverter. The generator side converter converts the generator AC power into DC power and the line side inverter converts the DC power into an AC power for injection into the utility grid.

(10) FIG. 2A shows an example of a power system **200** of a wind turbine **100** according to an embodiment. The power system comprises a generator or power source **201** and a power converter **202**. The power converter **202** comprises a machine side converter **203**, a line side inverter **204**. The power converter **202** may further comprise a DC-link **205** and a resistor **207** connected with a controllable switch **206**. The resistor and switch forms a power dissipation device, also known as a chopper **209**, for dissipating active power. The DC-link **205** comprises one or more DC-link capacitors which are charged by the DC output current from the generator side converter **203** and which supplies DC power to the line side inverter **204**. The output AC current from the line side inverter **204** may be supplied via output inductors **206** and possibly via a wind turbine transformer **208** to the grid or power line **220**. In this example, the output AC current is a 3-phase current output. Harmonic filter capacitors **216** may be arranged between the conductors of the output, which together with the inductors **206**, forms a harmonic filter which converts the square wave voltage signals from the line side inverter **204** to voltage sinusoidal signals.

(11) The power line **220** may be a medium voltage power bus which receives power from other wind turbines **100**. The power line **220** may be connected with a high voltage network, e.g. via further transformers. Thus, the power line **220** and one or more power systems **200** of corresponding wind turbines constitutes a wind power plant or park arranged to supply power to a utility grid for distribution of electrical power. The power line **220** and the high voltage network is commonly referred to as a power grid herein.

(12) The power converter **202** may be full-scale converter configured according to different principles including forced-commutated and line-commutated converters.

(13) The power system **200** is principally illustrated and therefore does not explicitly reveal that the system may be a three phase system. However, principles of the described embodiments apply both to single and multi-phase systems.

(14) The line side inverter **204** uses some variant of pulse width modulation (PWM) for converting the DC power into AC power. The control system **250** is used for controlling the modulation of the line side inverter **204** and for controlling the active power P and the reactive power Q generated by the line side inverter **204**.

(15) FIG. 2A shows that the grid voltage U_{grid} , here the voltage at the low voltage LV side of the transformer **208**, can be measured. The grid voltage U_{grid} can be used for determining a virtual synchronous machine angle θ_{VSM} (as described elsewhere) and for controlling the power output of the converter, based on determining the active power P_{grid} from grid voltage U_{grid} and grid current I_{grid} . The reactive power Q_{grid} may similarly be determined from U_{grid} and I_{grid} . Alternatively, the grid voltage U_{grid} may be measured on the high voltage HV side of the transformer and corrected based on the turns ratio of the transformer, or the internal voltage magnitude reference V_{qref} is used instead of the measured voltage U_{grid} . In an alternative, an internal voltage magnitude reference such as V_{qref} , V_{dref} or $V_{\alpha\beta ref}$ may be used for determining P_{grid} and consequently the synchronous machine angle θ_{VSM} . The grid current I_{grid} supplied to

the grid can also be measured.

(16) FIG. 2B shows an example of control components **260** arranged for controlling the generation of active power P_{grid} and reactive power Q_{grid} supplied to the grid at the power output **270** of the wind turbine **100**. That is, the control components **260** may be arranged for controlling the output active power P_{grid} and the output voltage magnitude at the low voltage side LV, alternatively for controlling the output active power P_{grid} and the output reactive power Q_{grid} at the low voltage side LV. The control components **260** such as the frame conversion unit **266** and the pulse width modulator **265** may form part of the control system **250** or receive control signals from the control system **250**.

(17) References for the active and reactive power may be received from a power plant controller, PPC, or a grid operator, TSO, or determined from active and reactive power references, e.g. from the grid operator.

(18) The active power, P_{grid} , is controlled via the virtual synchronous machine angle θ_{VSM} . In short, the synchronous machine angle acceleration (the double-time derivative of θ_{VSM}) corresponds to the difference between a power reference P_{ref} for a desired power output of the wind turbine and a grid power P_{grid} supplied by the wind turbine to a power grid. The synchronous machine angle θ_{VSM} is determined according to a grid forming inverter scheme such as a virtual synchronous machine control scheme. A grid forming inverter scheme models the inherent rotating mass inertia of synchronous generators. By modelling the inertia, the inverter may provide improved grid stability since the grid forming inverter model opposes changes in grid frequency. That is, an increase in the grid frequency causes an increase of the kinetic energy and rotation frequency of the inertia but with a response time determined by the inertia. Oppositely, a decrease in the grid frequency causes a decrease of the kinetic energy and frequency of the inertia but with a response time determined by the inertia. In a wind turbine, the increase or decrease of the kinetic energy of the modelled synchronous generator causes an increase or decrease of the kinetic energy of the rotor **102**.

(19) Examples for determining the synchronous machine angle θ_{VSM} is given elsewhere in this description.

(20) The synchronous machine angle θ_{VSM} may be used to transform the signals from the rotating DQ frame into a non-rotating frame such as the $\alpha\beta$ or abc frame, or vice-versa. Based on the synchronous machine angle θ_{VSM} and voltage magnitude reference V_{qref} , control signals for the desired active power and reactive power are determined.

(21) The synchronous machine angle θ_{VSM} is defined in a rotating DQ frame defined by the angular position θ_{VSM} and rotating with the frequency ω_{VSM} . Based on the synchronous machine angle θ_{VSM} , control signals, i.e. the angle of the modulation voltage signals for the pulse-width-modulator PWM, **265** are determined and transformed into a non-rotating frame such as the $\alpha\beta$ or abc frame. The modulation voltage reference signal controls the active and reactive power P_{grid} and Q_{grid} .

(22) The frame conversion unit **266** transforms the control signal from the DQ frame into the $\alpha\beta$ or abc frame and determines the sinusoidal voltage references for the PWM **265**. The frame converted output signals from the frame conversion unit **266** are converted by the pulse-width-modulator PWM, **265** into a modulation signal for the grid side converter **204** in order to generate the desired active power and reactive power and/or voltage magnitude.

(23) The voltage magnitude reference V_{qref} is provided as a reference for a desired grid voltage or a desired reactive power Q_{grid} to be generated by the inverter **204**. The voltage magnitude reference V_{qref} may be determined based on a difference between a reactive power reference Q_{ref} and an actual reactive power Q_{grid} delivered to the grid. Thus, the reactive power Q_{grid} to be generated by the line side inverter **204** can be controlled based on a voltage magnitude reference V_{qref} .

(24) The voltage magnitude reference V_{qref} may be defined in the DQ frame which rotates with

the rotational speed ω_{VSM} of the virtual synchronous machine, which in a steady state condition may equal the fundamental frequency such as 50 Hz of the AC grid voltage. The voltage magnitude reference V_{qref} , or a modification thereof as described in the following, may be converted from the DQ frame to the $\alpha\beta$ or abc frame and outputted from the frame conversion unit **266** as a control signal to the pulse-width-modulator PWM, **265** which determines the modulation signal for the grid side converter **204**.

(25) Depending on the governing grid codes for a wind turbine power plant, the power converters of wind turbines may be required to be operated as virtual synchronous machines for grid currents I_{grid} below a given overcurrent threshold I_{max} . According to some grid codes, this overcurrent threshold is suggested to be around 1.5 pu. If the overcurrent threshold is high, grid disturbances like phase jumps may lead to a high power peak or torque peak in the generator side and drive train and consequently cause an undesired increase in the mechanical load.

(26) In principle, when a too high peak of the generator power P_{gen} is detected, a reduction of the generator power P_{gen} could be invoked by reducing the power reference P_{ref} . However, due to the inertia of the virtual synchronous model such a reduction would only lead to a slow reduction of the generator power P_{gen} or generator torque T_{gen} .

(27) Examples below in FIGS. 3A-3B, FIG. 4, FIGS. 5 and 6A-6B presents solutions to this problem which are based on providing an adaptive grid overcurrent threshold I_{max} instead of a fixed overcurrent threshold. According to the solutions, the grid overcurrent threshold I_{max} is changed based on the generator power P_{gen} or generator torque T_{gen} in comparison with threshold values P_{max} , T_{max} . When the generator power P_{gen} or generator torque T_{gen} is above the threshold value, the overcurrent threshold I_{max} is decreased to limit the generator power P_{gen} and generator torque T_{gen} to reduce the mechanical load. For example, the grid overcurrent threshold I_{max} may be set to 1.5 pu as default and the threshold value of the reduced generator power maximum P_{max} may be set to 1.2 pu. In this case, if the generator power P_{max} goes above 1.2 pu, the overcurrent threshold I_{max} is immediately decreased to keep the generator power below 1.2 pu. After the generator power P_{gen} becomes below 1.2 pu, the overcurrent threshold is increased to 1.5 pu again.

(28) The solutions are based on a current limiting method which is based on the idea of inserting a virtual impedance Z_{vir} between the output of the line side inverter **204** and the grid or the grid connection such as the transformer **208**. The virtual impedance Z_{vir} is principally illustrated in FIG. 2A. The virtual impedance Z_{vir} may be determined for one or more phases. Here the virtual impedance Z_{vir} is determined for commonly for all phases, i.e. as the same virtual impedance. The virtual impedance could also be individually determined for each phase. By increasing the resistive and/or the reactive value of the virtual impedance Z_{vir} , the output current drawn from the output of the line side inverter can be reduced. As illustrated in connection with FIG. 4 and FIG. 5, the power value P_{vir} of the virtual resistance part R_{vir} of the virtual impedance Z_{vir} , i.e. the virtual power dissipation, is used in the swing equation of the virtual synchronous generator implementation to reduce acceleration of the virtual synchronous generator during overcurrent situations and thereby the output current I_{grid} of the line side inverter **204**.

(29) An overcurrent may be defined as an output current of the line side inverter which is greater than the overcurrent threshold value I_{max} . The overcurrent threshold value I_{max} may be a predetermined value. However, as will be illustrated a fixed I_{max} may led to high peaks in the generator power P_{gen} when the delivered grid current exceeds the overcurrent threshold value I_{max} . Therefore, the adaptive determination of I_{max} as illustrated in FIGS. 6A-6B is devised.

(30) FIG. 3A and FIG. 3B show examples of control systems **391** for determining the synchronous machine angle θ_{VSM} of the virtual synchronous generator.

(31) The synchronous machine angle θ_{VSM} is determined based on a virtual synchronous machine control concept which aims at generating a power response which corresponds to the power response from a real synchronous generator, including the inertia of the synchronous generator.

(32) The power error ΔP is determined as the difference $P_{ref}-P_d-P_{vsm}$, where P_{ref} is a power reference for the desired active power output of the wind turbine, P_d is a damping power determined according to the virtual synchronous model **301**, and P_{vsm} is a virtual grid power. The virtual grid power P_{vsm} is given as the sum of the measured grid power P_{grid} and the virtual power P_{vir} of the virtual resistor R_{vir} . Under steady state conditions, the value of the power error ΔP is zero.

(33) In response to a change in the grid power P_{grid} , e.g. due to an decrease in the grid voltage U_{grid} and a corresponding increase in the grid current I_{grid} , the power error ΔP becomes non-zero, which causes the angle θ_{VSM} to increase or decrease to reduce the power error ΔP . For example, in an overcurrent situation where P_{grid} decreases, e.g. due to a grid short-circuit, the power error value ΔP becomes positive and the synchronous machine speed ω_{VSM} will increase.

(34) Thus, in response to fluctuations in e.g. the grid power P_{grid} , the synthetic inertial response value becomes non-zero, which causes the virtual machine to either accelerate or decelerate to reach a new equilibrium condition. The new equilibrium is reached when the virtual grid power P_{vsm} is again following P_{ref} .

(35) The virtual synchronous machine control concept is utilized on the line side inverter **204** using a swing equation to calculate θ_{VSM} .

(36) FIG. 3A shows an example of an implementation of the virtual synchronous model **301**. The virtual synchronous model **301** includes a closed loop where the virtual synchronous machine rotational speed ω_{VSM} is determined based on a combination a feedback of a damping power P_d , a power reference P_{ref} for the desired active power output of the wind turbine, the active grid power P_{grid} supplied by the wind turbine to the grid via the power line **220**, the virtual resistor power P_{vir} and an inertial integration model **311**. The inertial integration model **311** is implemented as $1/(2Hs)$ where H is the inertia time constant and $1/s$ is the integration in s-domain. Accordingly, the combination of powers $P_{ref}-P_d-P_{grid}-P_{vir}=\Delta P$ is used as input for the inertial integration model **311**.

(37) Since the derivative of the synchronous machine rotational speed ω_{VSM} is proportional to the deviation between the power reference P_{ref} and the virtual grid power P_{vsm} , the integration of the difference ΔP gives the synchronous machine rotational speed ω_{VSM} .

(38) The grid power P_{grid} can be determined based on the measured grid voltage U_{grid} —or internal voltage references such as the voltage magnitude reference V_{qref} , or transformations thereof such as $V_{\alpha\beta ref}$ or V_{dqref} —and the measured grid current I_{grid} .

(39) The damping power P_d is determined as the difference between the rotational speed of the grid ω_L and the synchronous machine rotational speed ω_{VSM} multiplied with the damping factor D_p . The rotational speed of the grid ω_L , i.e. the grid frequency is determined from the measured grid voltage U_{grid} .

(40) The synchronous machine angle θ_{VSM} is determined based on an integration of the synchronous machine rotational speed ω_{VSM} according to ω_r/s , where ω_r is the rated synchronous generator speed.

(41) FIG. 3B shows an alternative virtual synchronous model **301** which is not based on a measured grid voltage U_{grid} , but instead the rotational speed of the grid ω_L is determined based on a high-pass filtering of the determined synchronous machine rotational speed ω_{VSM} , i.e. by determining the rotational speed of the grid ω_L as the output of the high-pass filter **313** which is arranged to filter the inputted synchronous machine rotational speed ω_{VSM} .

(42) Thus, the alternative virtual synchronous model **301** is not based on a measured grid voltage U_{grid} , but the damping part, e.g. the damping power P_d , is determined based on a high-pass filtering **313** of the synchronous machine rotational speed ω_{VSM} .

(43) In general, the virtual synchronous model **301** determines the angle of the virtual machine θ_{VSM} based on the combination of powers P_{ref} , P_d , P_{grid} , P_{vir} , the inertial integration model **311**, e.g. implemented as $1/(2Hs)$ and a feedback of the damping power P_d determined based on ω_{VSM}

and an integration of ω_{VSM} . In other words, the synchronous machine rotational speed ω_{VSM} and the synchronous machine angle θ_{VSM} are determined so that they are indicative of a integrated deviation between a power reference P_{ref} for a desired power output of the wind turbine and the virtual grid power P_{vsm} .

(44) The control systems **391** can be implemented based on power values P_{ref} , P_d , P_{grid} , P_{vir} but may equivalently be implemented based on corresponding torque values T_{ref} , T_d , T_{grid} , T_{vir} based on the relationship where power equals torque times rotation frequency, e.g. the synchronous machine rotational speed ω_{VSM} .

(45) FIGS. **4** and **5** show examples of determining the virtual impedance Z_{vir} and feeding the virtual resistor power P_{vir} into the virtual synchronous model **301**.

(46) A current magnitude I_{m1} or I_{m2} is determined based on the grid current I_{grid} . The current magnitude I_{m1} or I_{m2} is determined so that the current magnitude represents the instantaneous current magnitude of the current. In this context, instantaneous may be the actual current amplitude per sample, or an average obtained over two or more samples, such as 10 samples.

(47) The current magnitude I_{m1} , I_{m2} may be determined based on the current components I_a , I_b , I_c in the reference frame, i.e. the current amplitudes of the available phases, based on the current components I_d and I_q (FIG. **4**) in the rotation dq-frame and/or based on the current components I_α and I_β (FIG. **5**).

(48) The current components I_d , I_q are obtained by transforming the current signals I_a , I_b , I_c of the three-phase current signal I_{grid} from the measurement frame abc to the DQ frame which rotates with the virtual synchronous angular frequency ω_{VSM} . The abc/DQ transformation is performed based on the virtual synchronous machine angle θ_{VSM} . The abc/DQ transformation is the projection of the current quantities I_a , I_b , I_c —or other quantities such as voltages—onto the two-axis frame DQ which rotates according to the angle θ_{VSM} .

(49) The current components I_α , I_β are obtained by transforming the current signals I_a , I_b , I_c of the three-phase current signal I_{grid} from the measurement frame abc to the $\alpha\beta$ frame which is stationary. The abc/ $\alpha\beta$ transformation is the projection of the current quantities I_a , I_b , I_c —or other quantities such as voltages—onto the two-axis stationary frame $\alpha\beta$.

(50) The current magnitude I_{m1} may be determined as the square root of the squares of the current components I_α , I_β or the squares of the current components I_d , I_q . The current components I_α , I_β or I_d , I_q may be determined for each sample, possibly as an average over a few samples.

(51) Alternatively, the current magnitude I_{m2} may be determined based on the phase current magnitudes I_{ma} , I_{mb} , I_{mc} , e.g. determined as the current amplitude or peak current value within a period, of the current signals. Further, the current magnitude I_{m2} may be determined as the maximum of the determined current magnitudes I_{ma} , I_{mb} , I_{mc} , or in other ways dependent on the current magnitudes.

(52) If a fast response is needed, i.e. if a fast compensation via the virtual impedance Z_{vir} is needed, the first alternative of the current magnitude I_{m1} may be preferred, due to the faster computation of I_{m1} compared with I_{m2} .

(53) If a better performance in unbalanced situations is needed, the second alternative of the current magnitude I_{m2} may be preferred. In unbalanced cases, the magnitude of one phase, for example I_{ma} , may be much larger than other phases (I_{mb} and I_{mc}), consequently I_{m2} would be larger than I_{m1} . So, If I_{m2} is selected to limit the current, a suitable larger virtual impedance may be inserted.

(54) The virtual impedance value Z_{vir} is determined by initially determining the virtual resistance part R_{vir} of the virtual impedance Z_{vir} . The determination of the reactive part X_{vir} of the virtual impedance Z_{vir} is determined based on the initially determined virtual resistance R_{vir} , but the determination of X_{vir} is optional since the method also works without determining the reactive part X_{vir} . Advantageously, X_{vir} may provide more independent active and reactive power control loops during overcurrent conditions.

(55) The virtual resistance R_{vir} is determined based on the current difference or current error I_{err} ,

i.e. the difference between the current magnitude I_{m1} or I_{m2} and the overcurrent threshold value I_{max} . Consequently, the virtual reactance X_{vir} and the virtual impedance Z_{vir} is also determined based on the current difference I_{err} . When the current magnitude I_{m1} , or I_{m2} is lower than I_{max} , the current difference I_{err} becomes negative and the virtual impedance Z_{vir} is reduced (if not already equal to zero) until it is saturated to the zero value. Accordingly, in normal operation Z_{vir} and P_{vir} equals zero values.

(56) In this example, the virtual resistance R_{vir} is determined using a controller. For example, the controller may be an integral controller (I-controller), proportional controller (P-controller) or a proportional-integral controller (PI-controller). Accordingly, the control algorithm of the controller may include a proportional part which determines a value which is proportional with the current difference I_{err} and/or an integral part which determines a value which is proportional with a time integral of the current difference I_{err} . Accordingly, the virtual resistance R_{vir} is obtained based the values determined by the P-controller and/or the I-controller, or the PI-controller.

(57) The integral part of the controller comprises an integral gain K_i which may be determined dependent on the current difference I_{err} , such as the sign of the current difference I_{err} . For example, in order to achieve a fast control action when the current difference I_{err} is positive, i.e. when the current magnitude I_{m1} , I_{m2} is greater than I_{max} , the integral gain K_i may be set to a predetermined relative larger gain K_{i1} . For negative current differences I_{err} , the integral gain K_i may be set to a predetermined relative smaller gain K_{i2} to achieve a more stable response during unbalanced situations, where I_{m1}/I_{m2} may oscillate. Furthermore, the integral gain K_i could also depend on the magnitude of the positive or negative difference I_{err} , i.e. by having predetermined gain factors K_i for different signs and magnitudes of the current error I_{err} . Therefore, the integral gain K_i may be determined from two or more predetermined gains. In the above example with two gains K_{i1} and K_{i2} , K_{i1} is greater than K_{i2} . Thus in an example, when $I_{err} > 0$, $K_i = K_{i1}$ is selected to limit the grid current fast. When $I_{err} < 0$, $K_i = K_{i2}$ is selected to have a stable response for the unbalanced cases.

(58) The determined virtual resistance value R_{vir} may be limited by the limiter according to a lower minimum limit R_{min} and/or an upper maximum limit R_{max} to avoid determination of negative resistance values and/or very large values. The output from the limiter is referred to as R_{vir_lim} .

(59) As illustrated in FIG. 4, the virtual reactance value X_{vir} may be determined based on the determined virtual resistance R_{vir} or the limited virtual resistance R_{vir_lim} . For example, the virtual reactance value X_{vir} may be determined dependent on a predetermined factor " K_x/r " which determines a ratio between the virtual reactance value X_{vir} and the virtual resistance value R_{vir} . The K_x/r factor may be an empirically determined factor, e.g. determined based required damping and grid conditions. For example, the K_x/r factor could be a value in the range from zero to ten.

(60) The voltage ΔV_{dq} over the virtual impedance Z_{vir} is determined as shown in FIG. 4 as $\Delta V_d = R_{vir_lim} \times I_d - X_{vir} \times I_q$ and $\Delta V_q = R_{vir_lim} \times I_q + X_{vir} \times I_d$. For that purpose, the grid current I_{grid} is transformed into I_{dq} with components I_d , I_q in the DQ frame and the impedance voltage is determined in the DQ frame. If X_{vir} is not determined, X_{vir} is simply set to zero in the equations.

(61) In the example in FIG. 5, the voltage $\Delta V_{\alpha\beta}$ over the virtual impedance Z_{vir} is determined based on the grid current I_{grid} transformed into $I_{\alpha\beta}$ in the $\alpha\beta$ frame according to the Clarke transformation. The voltage may be determined as $\Delta V_\alpha = R_{vir_lim} \times I_\alpha + (X_{vir}/\omega_{VSM}) \times d(I_\alpha)/dt$ and $\Delta V_\beta = R_{vir_lim} \times I_\beta + (X_{vir}/\omega_{VSM}) \times d(I_\beta)/dt$ (not shown in FIG. 5).

(62) Alternatively, as shown in FIG. 5, in order to avoid the derivative part, only the virtual resistance R_{vir} or the limited resistance R_{vir_lim} is determined and the voltage over the virtual resistance is determined as $I_{\alpha\beta} \times R_{vir_lim}$ or $I_{\alpha\beta} \times R_{vir}$.

(63) As another alternative the voltage ΔV_{abc} over the virtual impedance Z_{vir} may be determined by multiplying the current components I_a , I_b , I_c with the resistance and/or reactance components of the virtual impedance Z_{vir} .

- (64) In general as illustrated with examples in FIG. 4 and FIG. 5, the virtual voltage $\Delta V_{\alpha\beta}$, ΔV_{dq} over the virtual impedance Z_{vir} is determined based on the virtual impedance value Z_{vir} and the grid current I_{grid} , or a transformation thereof.
- (65) In FIG. 4, the virtual voltage ΔV_{dq} is combined, here subtracted, with the voltage reference V_{dqref} to generate the modified voltage reference V_{dq_m} .
- (66) The virtual power P_{vir} of the virtual resistor R_{vir} is determined as $P_{vir} = 1.5 \times \text{Im}\{ \text{circumflex over ()} \}^2 \times R_{vir}$ or $1.5 \times \text{Im}\{ \text{circumflex over ()} \}^2 \times R_{vir}$.
- (67) The virtual grid power P_{vsm} is determined as the sum of the virtual power P_{vir} and the grid power P_{grid} which may be determined based on U_{grid} and I_{grid} .
- (68) Alternatively, the virtual grid power P_{vsm} may be determined as $P_{vsm} = 1.5 \times (V_{\Delta ref} \times I_{\alpha} + V_{\beta ref} \times I_{\Delta})$ or $P_{vsm} = 1.5(V_{dref} \times I_d + V_{qref} \times I_q)$, where $V_{\alpha ref}$ and $V_{\beta ref}$ are the components of $V_{\alpha\beta ref}$, and where V_{dref} and V_{qref} are the components of V_{dqref} . Accordingly, the virtual grid power P_{vsm} may alternatively be determined based on the voltage magnitude reference V_{qref} and the grid current I_{grid} , or transformations thereof according to the equations.
- (69) Thus, in general the virtual grid power P_{vsm} may be determined based on the virtual resistance R_{vir} and the grid current I_{grid} or transformations therefrom, or based on the voltage references (V_{dqref} , $V_{\alpha\beta ref}$) and the grid current I_{grid} or transformations therefrom.
- (70) As described in connection with FIGS. 3A and 3B, based on the deviation ΔP between the power reference P_{ref} for a desired power output of the power generating unit and the virtual grid power P_{vsm} combined with a damping power (P_d), the virtual synchronous machine rotational speed ω_{VSM} and/or the synchronous machine angle θ_{VSM} is determined by the virtual synchronous model **301**, where the derivative of the synchronous machine rotational speed (ω_{VSM}) is indicative of the power deviation ΔP .
- (71) The voltage reference V_{abc} for controlling the line side inverter **204** to generate the desired reactive power Q_{grid} is determined by the frame conversion unit **266** based on the virtual synchronous machine rotational speed or angle ω_{VSM} , θ_{VSM} , and a combination of the virtual voltage $\Delta V_{\alpha\beta}$, ΔV_{dq} and the voltage magnitude reference V_{dqref} , here the difference $V_{dqref} - \Delta V_{dq}$. Here it is noted that V_{dref} may be zero, so that V_{dqref} equals V_{qref} . The voltage magnitude reference V_{qref} is provided to achieve a desired reactive power Q_{grid} to be generated by the line side inverter **204**. Thus, the voltage reference V_{dqref} is the voltage reference in the DQ frame generated based on the voltage magnitude reference V_{qref} .
- (72) The virtual synchronous machine angle θ_{VSM} is used by the frame conversion unit **266** to perform the transformation of the voltage signal V_{dq_m} from the DQ frame into the voltage signal V_{abc} in the abc frame. The voltage signal V_{abc} is provided as a control signal to the pulse-width-modulator PWM, **265** in order to generate the active power P based on θ_{VSM} as described in connection with FIG. 2A. The virtual synchronous machine angle θ_{VSM} is determined as described in connection with FIGS. 3A-B and is used for determining the control signal V_{abc} via the frame conversion unit **266**.
- (73) Accordingly, if a positive current error I_{err} is generated, the virtual power P_{vir} determined from the virtual resistance R_{vir} is added to the grid power P_{grid} and results in an increased virtual grid power P_{vsm} . The increased virtual grid power P_{vsm} results in reducing the error ΔP and consequently decreasing of θ_{vsm} . This causes that the angle difference between V_{abc} and U_{grid} is decreased, which causes a decrease in the grid current I_{grid} supplied by the line side inverter **204** to the grid.
- (74) FIG. 5 shows an alternative example for determining and implementing the virtual impedance Z_{vir} and feeding the virtual resistor power P_{vir} into the virtual synchronous model **301**. Here the example, only shows determination of the virtual resistance R_{vir} .
- (75) In this example, the grid current I_{grid} , i.e. the current components I_a , I_b , I_c , is transformed from the measurement frame abc to $I_{\alpha\beta}$ in the stationary $\alpha\beta$ frame. The abc/ $\alpha\beta$ transformation transforms the three vector values in the measurement frame abc to two vector values in the $\alpha\beta$

frame.

(76) The voltage magnitude reference $V_{\alpha\beta\text{ref}}$ is the $\alpha\beta$ transformed version of $V_{dq\text{ref}}$, where $V_{dq\text{ref}}$ is the voltage reference $V_{dq\text{ref}}$ in the DQ frame generated based on the voltage magnitude reference $V_{q\text{ref}}$.

(77) The virtual synchronous machine angle θ_{VSM} is used by the frame conversion unit **266** for transforming the $V_{dq\text{ref}}$ from the DQ frame into $V_{\alpha\beta}$ in the $\alpha\beta$ frame. The voltage control signal $V_{\alpha\beta}$ is transformed by transform element **621** to the control signal V_{abc} in the measurement frame abc and is provided as a control signal to the pulse-width-modulator PWM.

(78) The determination of the virtual grid power P_{vsm} in FIG. 5 is equivalent to the determination of P_{vsm} in FIG. 4.

(79) The virtual voltages $\Delta V_{\alpha\beta}$ are combined, here subtracted, with the voltage reference $V_{\alpha\beta\text{ref}}$ to generate the modified voltage reference $V_{\alpha\beta_m}$.

(80) The modified voltage reference $V_{\alpha\beta_m}$ is transformed from the $\alpha\beta$ frame to voltage control signal V_{abc} in the measurement frame abc by the $\alpha\beta/abc$ transform element **621**.

(81) Accordingly, the response of the circuit in FIG. 5 due positive current error I_{err} is equivalent to the control response of the circuit in FIG. 4.

(82) FIG. 6A shows a solution for determining the grid overcurrent threshold I_{max} adaptively. The difference ΔP between the generator power P_{gen} and the maximum generator power $P_{\text{max-H}}$ is integrated by the control function **601**. The control function may include only an integration function $1/s$ or an integration function scaled by a gain factor K_i . Optionally, the control function may include a proportional term K_p which may be fixed or variable. The control function **601** may be configured so that the gain factor K_i —which is multiplied with the power difference ΔP or equivalently the integration term of the control function **601**—depends on the power difference ΔP , such as the sign of the power difference ΔP .

(83) For example, in order to achieve a fast control action when the power difference ΔP is negative, i.e. when the generator power P_{gen} is greater than $P_{\text{max-H}}$, the integral gain K_i may be set to a predetermined relative larger gain K_{i1} . For positive power differences ΔP , the integral gain K_i may be set to a predetermined relative smaller gain K_{i2} to achieve a more stable response. In this example, the integral gain factor K_i may be set to K_{i1} or K_{i2} , where K_{i1} is greater than K_{i2} . Thus, with $\Delta P < 0$, $K_i = K_{i1}$ is selected to reduce the overcurrent threshold I_{max} fast and with $\Delta P > 0$, $K_i = K_{i2}$ is selected to get back to the predefined and nominal maximum overcurrent threshold I_{max} robustly. The values of K_i may be determined according to stability criterial of the control system.

(84) The integral gain factor K_i may be determined in other ways dependent on the power difference ΔP . For example, K_i may be determined dependent on the magnitude of ΔP and/or the sign of ΔP . The gain factor K_i may be determined dependent on a function such as a continuous function which depends on ΔP or dependent on a look-up table that contains two or more values to be selected dependent on the power difference ΔP .

(85) The limit function **602** is configured to set an upper maximum grid current value $I_{\text{max-H}}$ and a lower maximum grid current value $I_{\text{max-L}}$ for limiting the output value of the control function **601** so that the maximum current value I_{max} provided by the limit function **602** is limited to a value between the lower maximum current value $I_{\text{max-L}}$ and the upper maximum current value $I_{\text{max-H}}$.

(86) With the solution based on the control function **601**, the maximum current value I_{max} will approach the lower maximum current value $I_{\text{max-L}}$, e.g. 1.2 pu. if the generator power P_{gen} is greater than a maximum generator power $P_{\text{max-H}}$. If the generator power P_{gen} is smaller than the maximum generator power $P_{\text{max-H}}$, the maximum grid current value I_{max} will approach the upper maximum current value $I_{\text{max-H}}$, e.g. 1.5 pu.

(87) FIG. 6B shows an alternative solution for determining the grid overcurrent threshold I_{max} adaptively. The hysteresis function **611** outputs either the the lower maximum grid current value $I_{\text{max-L}}$ if the measured generator power P_{gen} is greater than the maximum generator power P_{max}

or outputs the upper maximum grid current value $I_{\max-H}$ if the measured generator power P_{gen} is smaller than the low maximum generator power $P_{\max-L}$.

(88) A rate limit function **612** is provided to limit the rate of changes of the output from the hysteresis function so that the rate of change of changes between the upper maximum grid current value $I_{\max-H}$ and the lower maximum grid current value $I_{\max-L}$ is limited according to the slope of the rate limit function **612**.

(89) The slope R_i of the rate limit function **612** may depend on whether the generator power P_{gen} is greater than the maximum generator power $P_{\max}$ or is smaller than the low maximum generator power $P_{\max-L}$, e.g. so that the slope $R_i=R_{i1}$ is selected for $P_{\text{gen}}>P_{\max}$ and the slope $R_i=R_{i2}$ is selected for $P_{\text{gen}}<P_{\max-L}$, where $R_{i1}>R_{i2}$, to achieve a fast reduction of an overcurrent threshold $I_{\max}$ if required, and a robust return to the nominal maximum overcurrent threshold $I_{\max}$.

(90) The examples in FIG. 6A and FIG. 6B are based on the generator power P_{gen} , but can equivalently be based on the generator torque T_{gen} or other generator loads L_{gen} . Accordingly, the maximum generator power $P_{\max}$, a maximum generator torque $T_{\max}$ or other maximum generator load $L_{\max}$ are equivalent. Similarly, the low maximum generator power $P_{\max-L}$, a low maximum generator torque $T_{\max-L}$ or other low maximum generator load $L_{\max-L}$ are equivalent.

(91) FIG. 7 illustrates simulation results of an implementation with the adaptive determination of the maximum grid current value $I_{\max}$ (curves with reference **702**) and simulation results of an implementation with a fixed maximum grid current $I_{\max}$ (curves with reference **701**).

(92) At t_1 , a grid fault such as a phase jump occurs.

(93) In the generator power simulation, curve **701** shows a high peak force at t_2 (1.5 pu) and therefore a large generator load, whereas curve **702** shows that the adaptive determination of the maximum grid current value $I_{\max}$ limits the power peak significantly to 1.2 pu.

(94) The middle graph shows that the maximum grid current value ($I_{\max}$) is reduced from $I_{\max-H}$ to $I_{\max-L}$ at t_2 and increases back to $I_{\max-H}$ when P_{gen} falls below $P_{\max}$ shortly after t_2 .

(95) The lower graph shows that the grid current magnitude (I_{dq}) is limited to 1.5 pu in curve **701** for the fixed $I_{\max}$, while the current magnitude is reduced in curve **702** due to the adaptively determined reduction of $I_{\max}$.

Claims

1. A method for limiting generator loads in a wind turbine comprising a power converter with a line side inverter arranged to supply power to a power grid, the method comprising: determining a maximum grid current value based on a measured generator load and a limit function which sets an upper maximum grid current value and a lower maximum grid current value such that (i) the maximum grid current value moves towards the lower maximum current value if generator load is greater than a maximum generator load and (ii) the maximum grid current value moves towards the upper maximum current value if the generator load is smaller than the maximum generator load or is smaller than a low maximum generator load; determining a virtual impedance value based on a current difference between a current magnitude associated with a grid current and the maximum grid current value, where the virtual impedance value comprises a virtual resistance value; determining a virtual voltage drop over the virtual impedance value based on the virtual impedance value and the grid current; determining a virtual grid power based on (i) the virtual resistance value and the grid current or (ii) the grid current and a voltage magnitude reference for a desired grid voltage or a desired reactive power to be generated by the line side inverter; determining a virtual synchronous machine angle based on a power reference for a desired power output of the power generating unit and the virtual grid power; and determining a voltage reference for controlling the line side inverter to generate power based on the virtual synchronous machine angle, the virtual voltage drop, and the voltage magnitude reference.

2. The method according to claim 1, wherein determining the maximum grid current value

comprises determining a load difference between the measured generator load and the maximum generator load, integrating the load difference, and determining the maximum grid current value based on the integrated load difference subject to the upper maximum grid current value and the lower maximum grid current value of the limit function.

3. The method according to claim 2, wherein integrating the load difference comprises multiplying the load difference with a gain factor which depends on the load difference.
4. The method according to claim 1, wherein determining the maximum grid current value comprises selecting the lower maximum grid current value if the measured generator load is greater than the maximum generator load, selecting the upper maximum grid current value if the measured generator load is smaller than the low maximum generator load, and determining the maximum grid current value by limiting a rate of change of changes between the upper maximum grid current value and the lower maximum grid current value.
5. The method according to claim 1, wherein the virtual synchronous machine angle is determined so that a double derivative of the angle is indicative of a deviation between a power reference and the virtual grid power combined with a damping power.
6. The method according to claim 1, wherein the virtual resistance value is determined based on a control algorithm, where the control algorithm comprises a proportional part which determines a value which is proportional with the current difference or an integral part which determines a value which is proportional with a time integral of the current difference.
7. The method according to claim 1, wherein a virtual reactance value of the virtual impedance value is determined dependent on a predetermined factor defining a ratio between the virtual reactance value and the virtual resistance value.
8. The method according to claim 1, wherein the current magnitude is determined as the maximum current magnitude of the current magnitudes of current phases of the grid current, or as the magnitude of a current vector of the grid current or a transformation thereof.
9. The method according to claim 1, where a virtual synchronous machine rotational speed is determined based on a combination of a feedback of a damping power, the power reference, the virtual grid power, and an inertial integration model, where the synchronous machine angle is determined based on an integration of the synchronous machine rotational speed, and where the damping power is determined based on the virtual synchronous machine rotational speed.
10. The method according to claim 1, comprising: obtaining a network voltage at a connection point of the output of the power generating unit; determining a rotational speed of the grid based on the network voltage; and determining a damping power based on a virtual synchronous machine rotational speed and the rotational speed of the grid.
11. The method according to claim 1, comprising: determining a modified voltage reference based on the voltage magnitude reference and the virtual voltage drop; and determining the voltage reference based on the modified voltage reference.
12. The method according to claim 1, wherein the voltage magnitude reference is defined in a rotating direct quadrature frame.
13. The method according to claim 12, wherein the method comprises transforming the voltage magnitude reference to a transformed voltage magnitude reference in non-rotating target frame and wherein a modified voltage reference is determined by combining said transformed voltage magnitude reference with the virtual voltage drop.
14. A wind turbine control system arranged for limiting generator loads in a wind turbine comprising a power converter with a line side inverter arranged to supply power to a power grid, the wind turbine control system is arranged to perform an operation for limiting generator loads in the wind turbine, the operation comprising: determining a maximum grid current value based on a measured generator load and a limit function which sets an upper maximum grid current value and a lower maximum grid current value such that (i) the maximum grid current value moves towards the lower maximum current value if generator load is greater than a maximum generator load and

(ii) the maximum grid current value moves towards the upper maximum current value if the generator load is smaller than the maximum generator load or is smaller than a low maximum generator load; determining a virtual impedance value based on a current difference between a current magnitude associated with a grid current and the maximum grid current value, where the virtual impedance value comprises a virtual resistance value; determining a virtual voltage drop over the virtual impedance value based on the virtual impedance value and the grid current; determining a virtual grid power based on (i) the virtual resistance value and the grid current or (ii) the grid current and a voltage magnitude reference for a desired grid voltage or a desired reactive power to be generated by the line side inverter; determining a virtual synchronous machine angle based on a power reference for a desired power output of the power generating unit and the virtual grid power; and determining a voltage reference for controlling the line side inverter to generate power based on the virtual synchronous machine angle, the virtual voltage drop, and the voltage magnitude reference.

15. A wind turbine, comprising: a tower; a nacelle disposed on the tower; a rotor extending from the nacelle and have a plurality of blades disposed at a distal end thereof; a control system configured to perform an operation for limiting generator loads in the wind turbine, the operation comprising: determining a maximum grid current value based on a measured generator load and a limit function which sets an upper maximum grid current value and a lower maximum grid current value such that (i) the maximum grid current value moves towards the lower maximum current value if generator load is greater than a maximum generator load and (ii) the maximum grid current value moves towards the upper maximum current value if the generator load is smaller than the maximum generator load or is smaller than a low maximum generator load; determining a virtual impedance value based on a current difference between a current magnitude associated with a grid current and the maximum grid current value, where the virtual impedance value comprises a virtual resistance value; determining a virtual voltage drop over the virtual impedance value based on the virtual impedance value and the grid current; determining a virtual grid power based on (i) the virtual resistance value and the grid current or (ii) the grid current and a voltage magnitude reference for a desired grid voltage or a desired reactive power to be generated by a line side inverter; determining a virtual synchronous machine angle based on a power reference for a desired power output of the power generating unit and the virtual grid power; and determining a voltage reference for controlling the line side inverter to generate power based on the virtual synchronous machine angle, the virtual voltage drop, and the voltage magnitude reference.
